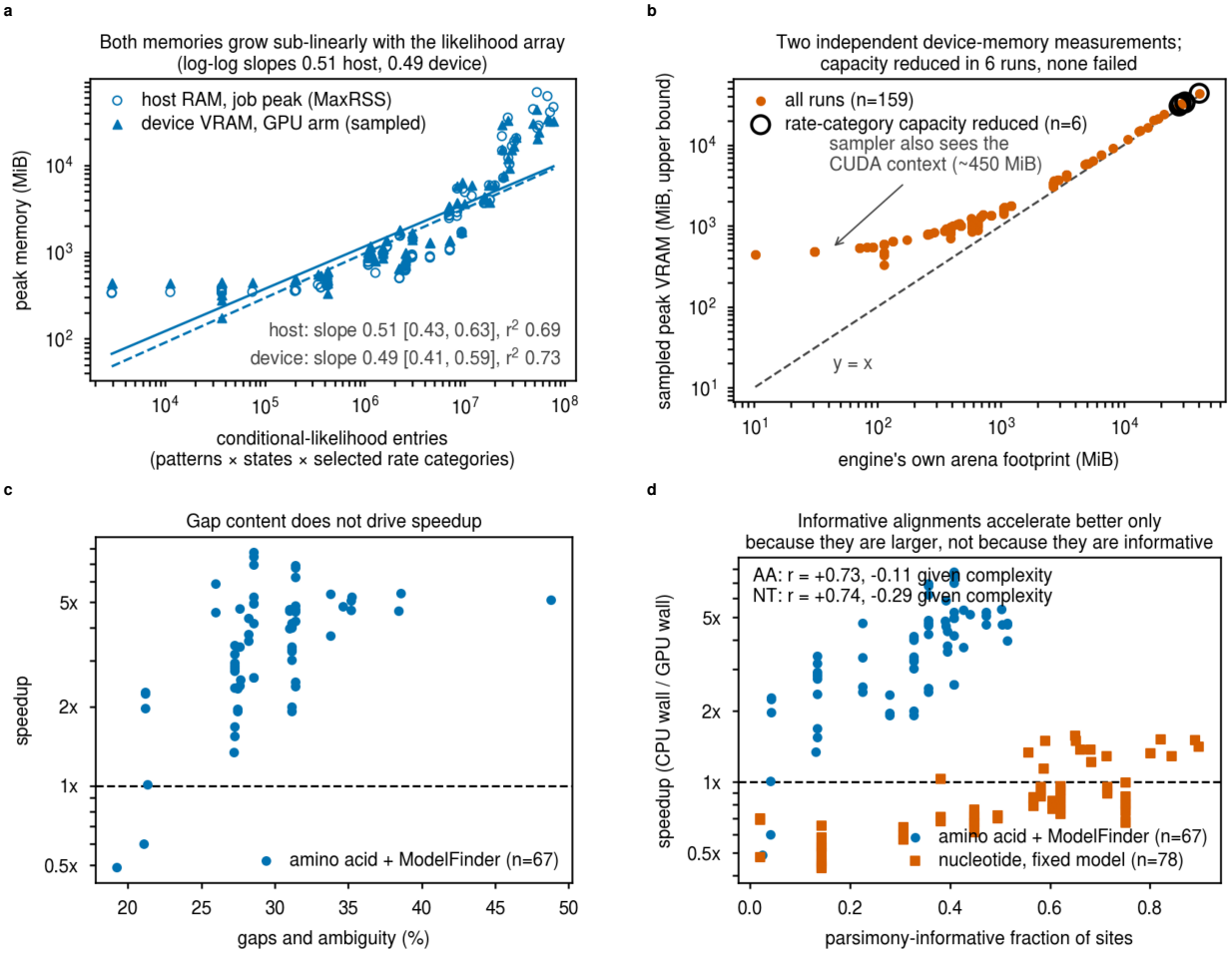
Figure 6


a
Both memories grow sub-linearly with the likelihood array
(log-log slopes 0.51 host, 0.49 device)
host RAM, job peak (MaxRSS)
device VRAM, GPU arm (sampled)
peak memory (MiB)
host: slope 0.51 [0.43, 0.63], r² 0.69
device: slope 0.49 [0.41, 0.59], r² 0.73
conditional-likelihood entries
(patterns × states × selected rate categories)
b
Two independent device-memory measurements;
capacity reduced in 6 runs, none failed
all runs (n=159)
rate-category capacity reduced (n=6)
sampler also sees the
CUDA context (~450 MiB)
y = x
sampled peak VRAM (MiB, upper bound)
engine's own arena footprint (MiB)
c
Gap content does not drive speedup
speedup
amino acid + ModelFinder (n=67)
gaps and ambiguity (%)
d
Informative alignments accelerate better only
because they are larger, not because they are informative
AA: r = +0.73, -0.11 given complexity
NT: r = +0.74, -0.29 given complexity
speedup (CPU wall / GPU wall)
amino acid + ModelFinder (n=67)
nucleotide, fixed model (n=78)
parsimony-informative fraction of sites